Whoever Picks the Tiles Decides: Selection Attacks on Efficient Whole-Slide Classifiers

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Abstract. Efficient whole-slide image (WSI) classifiers use a cheap model to pick a few tiles and encode only those with a pathology foundation model, reaching near full-slide accuracy at a fraction of the cost. We show that this tile selector is a single point of failure. An attacker who controls only the selector, and shows nothing but real, unmodified tiles from the slide, decides the diagnosis: on CAMELYON17, TCGA-RCC and TCGA-NSCLC, a compromised selector drives an ADMIL-style pipeline to the attacker's class on nearly every slide ($\kappa = -.24$; balanced accuracy .02 and .00). A larger tile budget does not help, while an honest but weak selector costs at most .01, so the failure is specific to adversarial selection. Debate-style presenters, with one advocate per diagnosis, do not solve the problem, because the true class's advocate is itself a single point of failure. We show that plain redundancy does: presenting the selector's top tiles in several interchangeable slots restores worst-case performance to .65–.74 on all three cohorts, with clean accuracy equal to the single selector and only K_c more tiles encoded per slide.

Keywords: Multiple-instance learning · Adversarial robustness · Efficient WSI inference

1 Introduction

Consider a lymph-node staging model deployed in a digital pathology workflow. To keep inference fast, the slide is first scanned at low magnification by a lightweight triage model, which hands the 10–20 most relevant tiles to a foundation-model classifier. Three facts make this hand-off a realistic point of attack. First, clinical digital-pathology platforms are modular: regulatory pathways such as predetermined change control plans let vendors add and update interoperable components without a new submission [23], so the selector and the classifier may come from, and be updated by, different parties. Second, healthcare creates concrete incentives to manipulate diagnostic AI, from billing to trial enrolment [5]. Third, a manipulated result is hard to catch: in a reader study, pathologists abandoned a correct assessment in 7% of cases when shown an erroneous AI recommendation [18], and here every tile the classifier relies on is a genuine region of the patient's slide. The same analysis also covers a faulty selector that fails in the worst possible way. [scene to confirm with clinical input]

Such pipelines are increasingly common. Encoding every tile of a gigapixel WSI with a pathology foundation model [3,15] dominates the cost of slide-level inference, so efficient methods select first and aggregate second [22,29,20,17]. In our experiments, a selector that passes 8–16 of 2,000–4,300 tiles (0.2–0.8%) to a multiple-instance learning (MIL) model stays within .014–.026 of full-slide MIL.

We ask how robust these pipelines are when the selector cannot be trusted. Our attacker is deliberately restricted: it never perturbs a pixel or a feature and shows no more tiles than the honest selector. It only chooses *which* real tiles of the slide are presented. Existing attacks and defences for MIL operate on pixels or features [6,28] and do not cover this threat, because every presented tile is genuine.

Contributions. (1) We formalise tile-selection attacks on efficient WSI classifiers, with a per-tile and a judge-aware adaptive attacker. (2) We show that the single-selector pipeline collapses under this attack on three cohorts and two tasks (metastasis staging, cancer subtyping), at every tile budget we test, although it is insensitive to honest selector quality. (3) We show that debate-style per-class advocates fail when the true class's advocate is compromised, and characterise when a presenter can survive one compromise (Prop. 1). (4) We show that simple redundant presentation restores robustness across all cohorts at negligible cost, and compare joint and isolated aggregation over 3 seeds $\times$ 5 folds.

2 Related Work

Efficient WSI inference. MIL on foundation-model features [7,14,19,11] reads every tile. ZoomMIL [22], RLogist [29], ADMIL [20] and EAGLE [17] select a few tiles with a cheap model first. Recent work studies how selection shifts the evidence MIL sees [27,9] and regularises attention to be faithful [12,4]; the selector is assumed to be trustworthy. **Adversarial robustness in pathology.** Attacks on MIL and pathology foundation models perturb tile pixels or features [6,28,13]. Our attacker uses only authentic tiles, which pixel-level defences cannot flag. **Debate and redundancy.** Debate [8,10], prover–verifier games [1] and Merlin–Arthur classifiers [24] let untrusted provers present evidence to a verifier; per-class advocates follow this idea. Isolate-then-aggregate defences judge each retrieved passage or image region separately [26,25]; we evaluate the WSI analogue alongside joint aggregation.

3 Method

Setting. A slide x with label y has tiles $\{t_1, \ldots, t_N\}$. A *presenter* fills slots $s = 1..m$; slot s shows a tile set S_s with $|S_s| \leq B_s$, each tile tagged with a one-hot slot flag. The judge J, a gated-attention MIL model [7] over the foundation-model features of the shown tiles, outputs class probabilities $p_J(\cdot \mid S_1, \ldots, S_m)$.

Threat model. The attacker compromises *one* slot of its choice and replaces S_s with any B_s tiles of the same slide; it cannot change the other slots, the flags or the features. We give it the true label and white-box access to J, the strongest

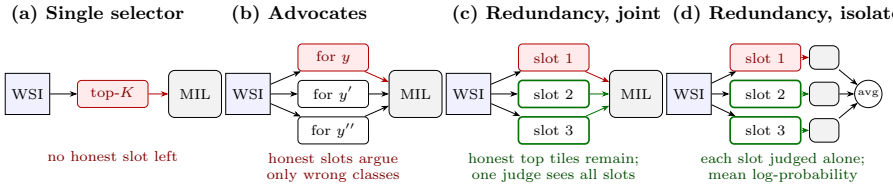


Fig. 1. Presenters under one compromised slot (red). (a) A compromised single selector decides alone. (b) With one advocate per class, compromising the true class's advocate leaves only slots that argue for wrong classes. (c, d) With redundant slots that each show part of the selector's top tiles, honest evidence for the true class remains (green), aggregated jointly (c) or slot by slot (d).

setting. *Per-tile attack*: for each wrong class c, show the B_s tiles with the highest single-tile $\log p_J(c)$ under slot s's flag. *Adaptive attack*: build S_s greedily from the top-128 per-tile candidates, adding at each step the tile that minimises $p_J(y)$ given the honest slots. We report the worst case per slide over slots, target classes and both attacks.

Presenters. *Single selector* (ADMIL-style [20]): one slot with the top-K tiles by the attention of a cheap scorer, an MLP on low-magnification EfficientNet-B0 [21] features distilled from a full-slide CONCH [15] ABMIL; $K = CK_c$ with $K_c = 4$. *Advocates*: one slot per class c with the K_c tiles of largest cheap-scorer margin for c (stage-specific spatial rules on CAMELYON17; see supplement). *Advocates + selector slot*: the advocates plus one unlabelled slot with the selector's top K_c or K tiles. *Redundancy*: $m = C{+}1$ interchangeable slots that receive the selector's top mK_c tiles round-robin, aggregated either *jointly* by one judge over all slots or *isolated*, by judging each slot alone and averaging log-probabilities [26].

Judge training. Multi-slot judges are trained on honest presentations and on presentations with one slot replaced by the per-tile attacker, for three rounds, over every (slot, target class) pair available to the attacker. The single selector's judge is the full-slide model.

When can a presenter survive one compromise? Call a slot a *witness* for (x, y) if it is honest and its tiles are chosen without regard to any wrong class: the advocate of y, the selector slot, or a redundant slot.

Proposition 1. *(i) The single selector is robust on (x, y) exactly when no set of at most K tiles of the slide leads J to a wrong class. (ii) For advocates, compromising the advocate of y removes every witness; all remaining slots are chosen to support wrong classes. (iii) Advocates + selector slot and redundancy with $m \geq 2$ slots retain at least one witness under any single compromise. With isolated aggregation, the presenter is robust on (x, y) exactly when, for every slot s and wrong class c, $\sum_{s' \neq s} \left[\log p_J(y \mid S_{s'}) - \log p_J(c \mid S_{s'})\right] > \max_{|S| \leq B_s} \left[\log p_J(c \mid S) - \log p_J(y \mid S)\right]$.*

Proof sketch. (i) The attacker controls the only slot and may show any $\leq K$ tiles, so it wins exactly when a misleading set exists. (ii) Advocate $c \neq y$ ranks tiles by their margin for c, so after the advocate of y is replaced, every slot is chosen to support a wrong class. (iii) The selector slot and redundant slots are filled by the class-agnostic selector, so with $m \geq 2$ at least one of them survives a single compromise. With isolated aggregation the prediction is $\arg\max_c \sum_s \log p_J(c \mid S_s)$; the attacker maximises its own slot's term, which gives the stated inequality. □

Part (i) explains the collapse, and its premise is a property of the tissue rather than of the model.

Why real tiles suffice: the biology. The attacker needs no fabricated evidence because diagnostic categories overlap at the scale of a tile. *Renal cell carcinoma:* clear-cell RCC frequently contains eosinophilic and granular areas that resemble chromophobe RCC, as well as papillary or pseudopapillary architecture that resembles papillary RCC; papillary RCC with eosinophilic cells in turn resembles both. *Lung cancer:* solid adenocarcinoma without visible glands is hard to separate from squamous carcinoma on morphology alone, keratinisation in squamous carcinoma can be focal, and mixed adenosquamous tumours contain both, which is why immunostains (TTF-1, p40) are used in ambiguous cases. *Lymph-node staging:* the pN category is defined by the *size* of the largest tumour deposit (isolated tumour cells $\leq$0.2 mm, micrometastasis $\leq$2 mm, macrometastasis >2 mm), a global spatial property that no handful of tiles can establish: tiles from the edge of a macrometastasis look like a micrometastasis, and a positive node is mostly tumour-free tissue. A selector that chooses which tiles to show can therefore make an honest slide look like another diagnosis using only tissue that is truly there. The data agree: scoring every tile on its own with the full-slide CONCH model, 96.7% of TCGA-RCC slides contain at least K=12 tiles assigned to a different subtype, and 99.2% of TCGA-NSCLC slides contain at least K=8 (median 24% and 29% of a slide's tiles), so the premise of Prop. 1(i) fails on nearly every slide. The pairs follow the morphology (Fig. 3d): 98% of clear-cell slides contain at least K tiles read as papillary, and 97–98% of chromophobe slides contain at least K tiles read as clear-cell or as papillary. [biological claims to be confirmed by the pathologist co-author]

Part (ii) explains why advocates fail exactly where it matters. Part (iii) is the structural property that redundancy provides; whether the judge follows the remaining witness is what our worst-case evaluation measures.

4 Experiments

Data. CAMELYON17 [2] (500 lymph-node slides, 4 pN stages, leave-one-hospital-out, quadratic-weighted κ), TCGA-RCC (882 slides, 3 subtypes) and TCGA-NSCLC (1,041 slides, 2 subtypes) [16] (5-fold patient-level CV, balanced accuracy). Tiles are encoded at 20× with CONCH and at low magnification with EfficientNet-B0. We report the mean over 5 folds, then mean ± sd over 3 seeds; paired 95% bootstrap CIs are over the 15 fold-seed runs.

Table 1. Clean and worst-case performance under one compromised slot. κ (CAMELYON17) or balanced accuracy (TCGA); worst case over the slot the attacker takes and the attack (adaptive for multi-slot presenters). Mean$_{\pm sd}$ over 3 seeds. Full-slide CONCH MIL: .890 / .933 / .923.

	CAMELYON17		TCGA-RCC		TCGA-NSCLC	
presenter	clean	worst	clean	worst	clean	worst
single selector	.873±.021	−.241±.009	.907±.019	.022±.002	.909±.007	.003±.000
budget $K+K_c$	.883±.013	−.233±.008	.908±.018	.028±.001	.911±.008	.006±.002
budget $2K$	.891±.011	−.212±.013	.909±.016	.036±.001	.914±.006	.008±.002
weak honest scorer (10% data)	.867±.025	–	.899±.011	–	.901±.003	–
advocates	.836±.010	.589±.024	.924±.011	.433±.003	.865±.004	.153±.003
independent scorers	.840±.006	.677±.018	.915±.017	.675±.029	.872±.005	.505±.048
advocates + selector slot (K_c)	.854±.004	.658±.009	.933±.013	.661±.027	.898±.003	.618±.007
advocates + selector slot (K)	.879±.012	.705±.021	.924±.011	.713±.026	.905±.001	.665±.018
redundancy, isolated	.836±.011	.748±.035	.899±.004	.706±.027	.886±.004	.464±.006
redundancy, joint	.869±.008	.654±.038	.915±.002	.739±.014	.914±.004	.725±.003

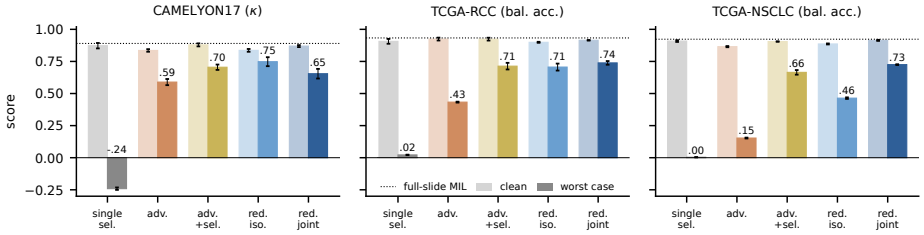


Fig. 2. Clean (light) and worst-case (dark) performance of the main presenters (Tab. 1); dotted: full-slide MIL. Only redundancy and the advocates with a selector slot stay well above chance on every cohort; joint redundancy does so at the clean accuracy of the single selector.

The single selector collapses (Tab. 1, Fig. 2). Compromised, it predicts the attacker's class on almost every slide: $\kappa = -.24$ on CAMELYON17 and .02/.00 balanced accuracy on RCC/NSCLC, consistently across folds and seeds. A larger budget does not help (−.21/.04/.01 at $2K$ tiles). At the same time the pipeline is insensitive to honest selector quality: a cheap scorer distilled from only 10% of the training slides costs $\leq$.01. The vulnerability is therefore specific to adversarial selection. [Running: the same attack against mean-pooling, max-pooling, TransMIL-style and DSMIL-style judges (Tab. 2), and a label-free, judge-free attacker.]

Advocates fail at the true advocate. As predicted by Prop. 1(ii), compromising the advocate of y leaves the judge with evidence for wrong classes only, and the worst case falls to .59/.43/.15. Training an independent scorer for each advocate, so that an attacker would have to compromise them separately, does not change this (.68/.68/.51 even under the weaker per-tile attack).

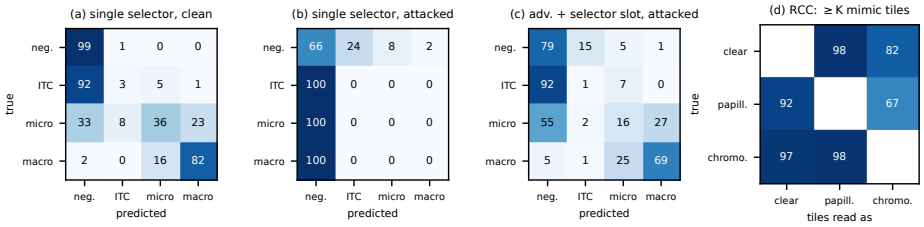


Fig. 3. Clinical impact. (a–c) CAMELYON17 pN-stage confusion (% of slides per true stage, 5 folds × 3 seeds). Compromised, the single selector reports every positive node as negative (b); with a selector slot beside the advocates, most macrometastases are still found (c). (d) TCGA-RCC: % of slides of each subtype that contain at least K=12 tiles the full-slide model reads as another subtype.

What the attack does clinically. On CAMELYON17 the worst-case attacker hides metastases rather than inventing them (Fig. 3a–c). The compromised single selector reports *every* positive slide (ITC, micro- and macrometastasis) as negative, in every fold and seed, while the honest selector misses 17.6% of micro- and macrometastases. With one compromised slot, advocates + selector slot misses 26.5% and keeps 69% of macrometastases at the correct stage. Every presented tile is genuine tissue, so a pathologist who inspects the evidence sees nothing wrong, only node tissue without tumour.

Adaptive evaluation matters. The judge-aware adaptive attack is stronger than the per-tile attack for every multi-slot presenter, by .02–.38 (e.g. advocates on NSCLC: .53 per-tile vs. .15 adaptive; joint redundancy on RCC: .86 vs. .74). Evaluating multi-slot defences against fixed attacks alone would therefore overstate their robustness.

Redundancy restores robustness. Joint redundancy keeps .654/.739/.725 under the strongest attack, up from −.241/.022/.003 for the single selector, while matching its clean accuracy (.869/.915/.914 vs. .873/.907/.909). Adding a selector slot to the advocates reaches a similar worst case (.705/.713/.665) with the highest clean accuracy on CAMELYON17 and RCC; joint redundancy is significantly more robust on NSCLC (+.060, 95% CI [.042, .080]). Isolated aggregation gives the best worst case on CAMELYON17 (.748) but drops to .464 on NSCLC. Prop. 1(iii) explains the ordering: the C honest slots must jointly outweigh the log-odds that one adversarial slot of K_c mimic tiles can reach, and their number falls from four (CAMELYON17, .748) to three (RCC, .706) to two (NSCLC, .464). A joint judge instead weighs the slots against each other and is not bound by this sum, so joint aggregation is the more reliable choice across cohorts. [bounded per-slot aggregation (clipped log-probability, majority vote) running] We therefore recommend joint redundancy as the default: it needs no class-specific selection rules and no change to the selector.

Implementation. Cheap scorers and judges are trained with AdamW, one slide per step; multi-slot judges get a 10-epoch warm start and three adversarial

rounds of 5 epochs (details in the supplement). Code will be released upon acceptance.

Cost. Redundancy encodes $(C{+}1)K_c$ instead of CK_c tiles (20/16/12 vs. 16/12/8, out of thousands). Encoding the selected tiles takes <0.3 s of a 5.5–7.9 s pipeline that is dominated by reading the slide at low magnification, so the overhead is negligible.

5 Discussion and Conclusion

Efficient WSI classifiers concentrate trust in the tile selector, a component that is accurate when honest but decides the diagnosis when compromised. Because the attack uses only authentic tiles, it is invisible to pixel-level defences and to a reviewer who checks the presented evidence tile by tile; in lymph-node staging it silently turns positive nodes negative. The root cause is biological as much as computational: tumour subtypes share morphology at the scale of a tile, and nodal stage is defined by the extent of a deposit, which a handful of tiles cannot show.

Implications for deployment. Three lessons follow. (1) Robustness to an honest but weak selector says nothing about robustness to a compromised one: the same pipeline loses at most .01 with a poorly trained scorer and everything with an adversarial one. (2) Splitting evidence selection into per-class advocates, as debate-style designs do, does not remove the single point of failure; it moves it to the advocate of the true class. (3) Presenting the selector's top tiles in several interchangeable slots, and judging them jointly, removes it at the cost of K_c more encoded tiles per slide and with no change to the selector. Since slot assignment is round-robin, the change is a few lines in existing pipelines.

Our attacker controls one slot. Several colluding components, attackers without label or model access, and a prospective evaluation with pathologists reviewing presented evidence are natural next steps.

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